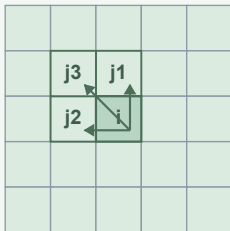


UPA Relative Geometry



Horizontal, vertical, and diagonal relations remain distinguishable

Key j1: $(\Delta r, \Delta c) = (0, +1)$

Key j2: $(\Delta r, \Delta c) = (+1, 0)$

Key j3: $(\Delta r, \Delta c) = (+1, +1)$

Separable Relative Bias

Relative Displacement

$$\Delta r = r_i - r_j, \Delta c = c_i - c_j$$

Separable Bias

$$B_h(\Delta r, \Delta c) = b_r^h(\Delta r) + b_c^h(\Delta c)$$

Attention Score

$$QK^T / \sqrt{d} + B_h(\Delta r, \Delta c)$$